

HarbourView Community Library Practice SBA

Full School-Based Assessment project brief

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: HarbourView Community Library

Sector: Education

Description of Project

HarbourView Community Library serves students and adults from nearby communities. Staff want to track members, books and loans more accurately, identify overdue items, analyse borrowing patterns and make library information easier to access.

Create a connected library project that manages lending data, analyses overdue charges, prepares member communication, presents library services and automates fine calculations.

How to use this practice SBA

- Read the whole project before starting any software task.
- Complete the sections in order so information remains consistent from one file to the next.
- The wording and data are original SPARK practice material. Your teacher's current assignment takes priority.
- The practice task headings follow the style of common CSEC school assignments. Current CXC limits should still be followed.

Contents

Section	Practice weighting
Database Management	20 practice marks
Spreadsheet	20 practice marks
Word Processing	20 practice marks
Web Page Design	10 practice marks
Problem-Solving and Programming	15 practice marks

Database Management - 20 practice marks

Task A - Tables

1. Create a database named HarbourViewLibrary_YourFullName.accdb.
2. Create the Member, Book and Loan tables using suitable field names, data types and primary keys.
3. Enter realistic records and apply the validation rules required by the project.
4. Create the required relationships and enforce the correct links between the tables.

Task B - Queries and Form

1. Create a query called Overdue Loans. Show loans returned after the due date or still overdue. Include member, book and days late.
2. Create a query called Member Loans. Join Member, Book and Loan and show all borrowing details sorted by LastName.
3. Create Member Information Form with Loan subform and clear borrowing headings.

Task C - Report

1. Create Overdue Report grouped by MemberType and showing count and total FineDue.

Spreadsheet - 20 practice marks

Task A - Calculations and Functions

1. Create LoansAnalysis_YourFullName.xlsx with worksheets named Loans and Dashboard. Enter or import the project data into Loans and format it as a clear data table.
2. Create the main calculations using efficient formulas. The SPARK reference uses formulas such as =MAX(0,F2-E2), =IF(G2>0,G2*H2,0) and =I2*(1-\$M\$2).
3. Use at least three suitable functions. Functions used in the SPARK reference include COUNTIF, AVERAGE, SUM, IF, MAX.

Task B - Analysis and Presentation

1. Sort and filter the data to answer this requirement: Show overdue loans only, then sort by DaysLate descending.
2. Create a summary showing Summary of number of loans by Category. Then create a suitable chart.
3. Create a linked total or key result on the Dashboard worksheet so that it updates when the detailed sheet changes.

Word Processing - 20 practice marks

Task A - Fillable Form

1. Create a document named Library Membership Form_YourFullName.docx. The form must collect the information needed for this project and include clear instructions.
2. Use suitable form controls. The SPARK reference uses: Full name text box; Member type drop-down list; Membership date picker; Terms check box.

Task B - Mail Merge

1. Create a professional main document called Overdue Notice. Use the project data as the mail-merge source.
2. Insert the following merge fields where they make sense: MemberName, BookTitle, DueDate, FineDue.
3. Complete the final merged output and include one relevant table, chart or other imported item from the connected SBA work where appropriate.

Web Page Design - 10 practice marks

Web Page Design

1. Create one web page for HarbourView Community Library. The page must be suitable for the intended audience and include a project logo.
2. Organise the page into these content sections: Welcome, Services, Borrowing Rules, Opening Hours, Contact.
3. Add at least two suitable hyperlink types and test them.
4. Add at least one relevant graphic and make sure the page remains readable at different screen sizes.
5. Save, test and prepare the final web-page evidence.

Problem-Solving and Programming - 15 practice marks

Task A - Problem Definition and Algorithm

1. Write a clear problem statement for the following task: Calculate the overdue amount for each library member and the total fines from all records processed.
2. Identify the inputs, processes and outputs, then write pseudocode containing at least one selection and one repetition statement.

Task B - Trace Table and Test Data

1. Create a trace table using at least three sets of test data. Include normal data, a boundary case and an invalid or unusual case where appropriate.

Task C - Program Implementation

1. Implement the algorithm in the programming language selected by your centre. The SPARK Pascal reference is saved as LibraryFines_YourFullName.pas.
2. Test the program and capture evidence that the expected and actual results agree.

Task D - Documentation

1. Create ProgramDocumentation_YourFullName.pdf containing the problem definition, algorithm, source code, trace table, test data and screenshots of program execution.

Submission checklist

- HarbourViewLibrary_YourFullName.accdb
- LoansAnalysis_YourFullName.xlsx
- Library Membership Form_YourFullName.docx
- Overdue Notice_YourFullName.docx
- One completed web page with its supporting files
- LibraryFines_YourFullName.pas
- ProgramDocumentation_YourFullName.pdf

Final reminder

Keep the same names, codes, dates, fees and rules across your database, spreadsheet, word-processing, web and programming work. A strong SBA reads like one connected project.